

Stephen Casse · ksh

Freelance DevOps / SRE Engineer — Verification · Automation

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Eleven years in critical banking production (Amundi, BPCE, Euler Hermes), plus the products I build and run myself — Stripe payments, PSD2 bank aggregation. I apply production engineering to AI systems: not the prompts, everything around them — executable proof, measured cost, audited security, a tested exit plan.

11 years

critical banking production

20,600 tests

on a SaaS I run myself:
payments, bank aggregation

**~1,200
servers**

self-service log consultation,
large account

1 standard

portable Apache stack, RHEL 6
→ 9.5, adopted in production

// EXPERIENCE

CS Consulting

Feb 2020 → present

DevOps & Platform Engineer — Founder · Paris · hybrid · independent

My own company: I design and run products end to end — architecture, specification, go-live, operations, support. Sole trader in 2020, incorporated in 2023.

- **From carbon paper to a multi-tenant SaaS:** Astre Solutions ran its purchase and delivery notes on carbon paper, in duplicate. I delivered a macOS document generation tool, synchronised and backed up off site. The same client then asked me to automate refrigerant-gas regulatory forms for Co&Sta, then the CRM, the scheduling, the field technicians — it became **Norteo**, now offered to other refrigeration companies with their agreement.
- **Norteo**, a multi-trade management SaaS I design and operate: 15 trades, 70 data models, white-label multi-instance architecture, JWT tenant isolation, TOTP MFA. Stripe subscriptions, bank aggregation through Bridge API v3 (PSD2-licensed aggregator) with automatic payment reconciliation. Regulatory documents generated automatically.
- What holds it together: **20,600 automated tests** — 19,000 unit (Vitest), 1,300 end-to-end (Playwright) — a blocking SonarQube quality gate before build, static security analysis, secret and dependency vulnerability scanning in CI. These products carry money paths: I do not see my job as writing code, but as **specifying what must be true and building the proof that it is**.

BPCE IT

Nov 2024 → present

DevOps Engineer, Build & Integration · Paris · hybrid · freelance

Turn an abandoned internal tool into a group-wide service, across an estate spanning four RHEL generations.

- **Logs for developers, without giving them production:** SSH access to production is denied to development teams — so no autonomous way to read an application log, and longer resolution times on every incident. I took over an outdated, unmaintained and insecure tool and turned it into a self-service log consultation service: one instance per server, a dedicated URL, LDAP/Kerberos authentication, browsing and download from the browser. The friction disappears, the security boundary stays intact. **Deployed on ~1,200 servers**, adopted by all fifteen business IT departments of the division and used by around 200 people across business IT and operations; presented to the executive committee, under review as a group product.
- **What makes that deployment possible:** an Apache stack compiled end to end — Apache, APR, OpenSSL, OpenLDAP, Expat — with no system dependency, runnable without root privileges, from RHEL 6 to RHEL 9.5. It installs on a bare Red Hat, coexists with an existing Apache or JBoss and survives an OS upgrade — **adopted as the production standard**. Delivery pipeline Bitbucket → Jenkins → Artifactory → XL Deploy.
- **Inventory derived from the CMDB:** the Ansible Automation Platform inventory was maintained by hand; it is now built dynamically from the CMDB with normalised environments — derived from the source of truth instead of being declared twice. Same principle for Java keystores and truststores.
- Daily run, 24/7 on-call rotation, application-domain lead with the business IT departments; team training (Ansible, Jenkins, Artifactory, Bitbucket) and group-wide Ansible Automation Platform presentation.

Amundi

Jan 2022 → Oct 2024

DevOps Engineer — application lead, deputy tech lead · Paris · hybrid · freelance

Application scope of a finance division — finance, employee savings, private banking: four to five packaged products, business stakeholders on the demand side, direct support to the team leader.

- **End-to-end SAS migration:** a strategic analytics platform reaching end of support — without an upgrade, the finance division would lose it. Six months from scoping to go-live, alongside a vendor consultant: study of the possible targets and sizing with the infrastructure teams, scoping of the business perimeter with users, provisioning, installation, network, LDAP and Active Directory integration, documentation, handover.
- Automated fail-over of application and database clusters during DR exercises: a manual procedure made **repeatable identically**, with no human input. Deployments Git → AWX → Ansible and Control-M “as code” (JSON/Jinja2 templates deployed by playbook). Daily application status collected and published by the pipeline instead of being checked by hand every morning.
- Oracle Financials & OBIEE migration from Solaris to SUSE; applications running on a Rancher Kubernetes platform driven by ArgoCD. Diagnosis of incidents across the whole scope, in particular those reported with no usable technical evidence; team meetings run in English; onboarding of newcomers and internal documentation.

Earlier: BPCE Lease (2020 → 2021) — migration of a critical reporting application to Azure, taken over and delivered single-handed after the project manager left · Amundi (2017 → 2020) — operations analyst, application production · Euler Hermes (2015 → 2017) — operations & datacenter technician.

// SKILLS

Automation & CI/CD

Ansible Ansible Automation Platform 2 AWX Jenkins
GitLab CI Bitbucket Artifactory XL Deploy / XL Release
ArgoCD Helm Kustomize

Cloud & IaC

Azure and Terraform (BPCE Lease migration 2021) XCP-ng
vSphere Docker Compose

Verification

Unit testing end-to-end testing Vitest Playwright
SonarQube static security analysis secret detection
vulnerability scanning

Observability

Prometheus Grafana Loki OpenTelemetry VictoriaMetrics
Centreon Zabbix AppDynamics Splunk

Languages

Bash Python jq YAML Jinja2 JSON Rust (basics)

Containers & orchestration

Docker Kubernetes K3s Longhorn MetalLB ingress-nginx
cert-manager Sealed Secrets

Systems

Red Hat (RHEL 6 → 9.5) SUSE Debian/Arch AIX Solaris
Windows Server Apache HTTPD JBoss Nginx

AI & agents

LLMOps Model Context Protocol (MCP) AI agents
LLM security per-model cost tracking Anthropic Claude

Security

SSL/TLS OpenSSL JKS keystores Venafi LDAP Infisical
Authentik Linux hardening

// PROJECTS

Homelab K3s, 100 % GitOps

homelab.scasse.com

3 Raspberry Pi 4, K3s, MetalLB, Longhorn, cert-manager, Sealed Secrets, Argo CD App-of-Apps (9 apps), 3 Ansible playbooks, Hugo docs deployed by GitLab CI. As of 2025: the infrastructure has since moved to an XCP-ng platform.

Skytnet — local-first AI cockpit

[Claude](#) · [MCP](#)

Personal assistant driven by Claude, run like a production application: zero-tolerance CI, deployment gated on green, OpenTelemetry observability, per-model cost tracking, security audits replayed on every change, quarterly recovery drill.

nalarch — open source

[Rust](#) · [MIT](#)

TUI package manager for Arch Linux; nothing runs without an approval screen.

// EDUCATION

2018 ITIL certification (level 1)

2017 École 42 Paris “Piscine” — C programming, peer-graded, no teachers

2016 AIX administration certification (level 1)

// LANGUAGES

French — native

English — professional — team meetings run in English